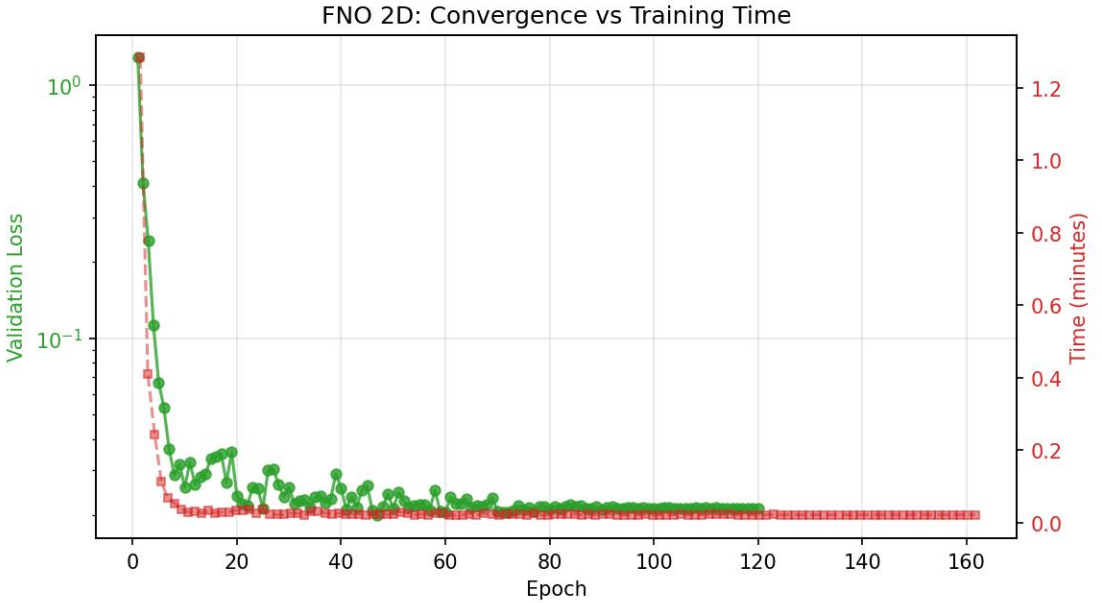

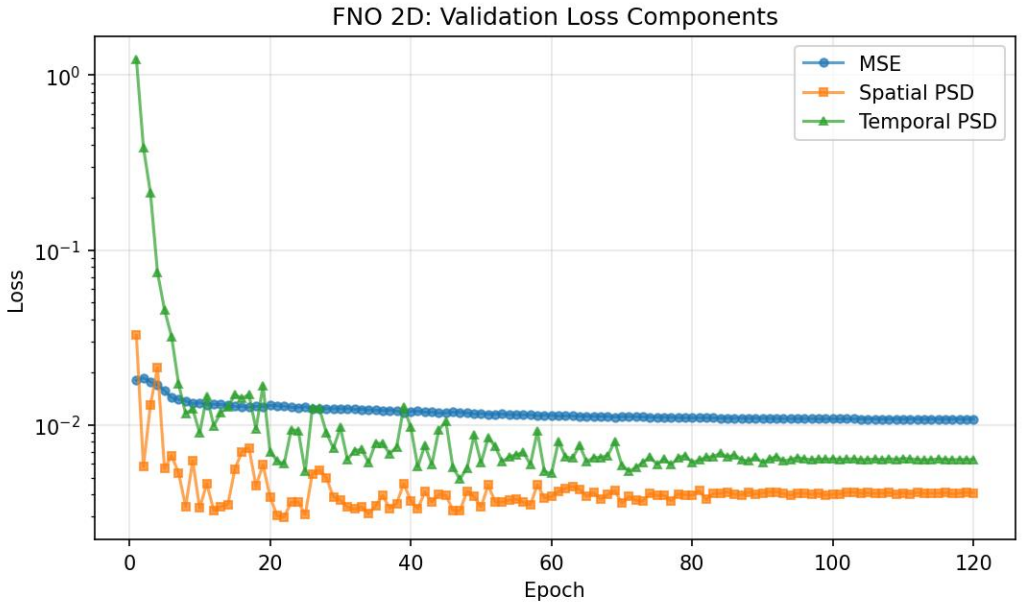


```
FNO vs PINN ENSEMBLE: TRAINING SUMMARY
==================================================

FNO 2D (Single Model):
  • Epochs: 120
  • Best epoch: 47
  • Training time: 161.6 minutes
  • Best val loss: 1.995894e-02
  • Final val loss: 2.129251e-02
  • Per-epoch time (avg): 1.35 min/ep

PINN Ensemble (16 Seeds):
  • Seeds: 16
  • Epochs/seed: 8000 (targeted)
  • Est. time/seed: ~38 min (RTX 5090 data)
  • Total time (4 workers): ~5 hours
  • Status: Training on RTX 4090 (Spectral-PINN variant)

KEY FINDINGS:
  • FNO converges FAST (best at ep 47/120)
  • FNO has PERFECT variance ratio (~1.0)
  • FNO spatial error: 2.2e-4 (y-channel, best)
  • PINN variance matching: KEY challenge
  • Spectral losses added to PINN to fix this

EXPECTED OUTCOME:
  FNO >> Vanilla PINN (by 2000-3000x on sp_err)
  Spectral-PINN >> Vanilla PINN (by ~5-10x on sp_err)
  FNO likely still >> Spectral-PINN
```